

L6: Import, Export ER-diagram

RECAP from Lab #5

- Insert Data

```
CREATE TABLE contacts(  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(255) NOT NULL,  
    email VARCHAR(320) NOT NULL);
```

```
CREATE TABLE contacts (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    first_name VARCHAR(50) NOT NULL,  
    last_name VARCHAR(50) NOT NULL,  
    fullname varchar(101) GENERATED ALWAYS AS  
        (CONCAT(first_name, ' ', last_name)),  
    email VARCHAR(100) NOT NULL);
```

Types of Files

The most common and efficient types of files to import/export with MySQL are:

- *.sql for full database dumps.
- *.csv for table data.

6.1 Import from *.sql

1. By script: Just execute the *.sql file.

2. By CMD:

- Create a blank new_database.
- `$ mysql -u <username> -p new_database < data-dump.sql`

1. By workbench:

- Click on **Data Import/Restore** on the left side of the screen.
- Select Import from **Self-Contained** File radio button (right side of screen)
- Select the path of .sql
- Click **Start Import** button at the right bottom corner of window.

MySQL Workbench

Local instance MySQL84 - W...

File Edit View Query Database Server Tools Scripting Help

SQL File 2* Administration - Data Import/Res... Dump20241211 (1)

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore**

INSTANCE

- Startup / Shutdown
- Server Logs

Administration Schemas

Information

No object selected

Data Import

Import from Disk Import Progress

Import Options

☐ Import from Dump Project Folder C:\Users\MUICT\Documents\dumps ...

Select the Dump Project Folder to import. You can do a selective restore.

Load Folder Contents

☒ Import from Self-Contained File C:\Users\MUICT\Documents\dumps\Dump20241211.sql ...

Select the SQL/dump file to import. Please note that the whole file will be imported.

Default Schema to be Imported To

Default Target Schema: lab7 New...

NOTE: this is only used if the dump file doesn't contain its schema, otherwise it is ignored.

Select Database Objects to Import (only available for Project Folders)

Imp...	Schema
--------	--------

Imp...	Schema Objects
--------	----------------

Dump Structure and Dat Select Views Select Tables Unselect All

Import Completed

Start Import

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 242	14:31:40	/*!40111 SET SQL_NOTES=@OLD_SQL_NOTES */	0 row(s) affected	0.000 sec
✓ 243	14:31:43	show databases	14 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Query Completed

6.1 Export to *.sql

1. By CMD:

- `$ mysqldump -u <username> -p database_name > data-dump.sql`

2. By workbench:

- Click on **Data Export** on the left side of the screen.
- Select objects to export.
- Select **Export to Self-Contained File**.
- Check **Include Create Schema** (option: if you want to load by execute file)
- Click **Start Export** button at the right bottom corner of window.

MySQL Workbench

Local instance MySQL84 - W... x

File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 2* Administration - Data Export x

MANAGEMENT

- Server Status
- Client Connections
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- Data Export**
- Data Import/Restore

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Information: No object selected

Local instance MySQL84
Data Export

Object Selection Export Progress

Tables to Export

Exp...	Schema
☐	chinook
☐	classicmodels
☐	l02_xxx_pokemon
☐	lab2
☐	lab4
☐	lab5
☒	lab6
☐	lab7
☐	mydb
☐	new_db
☐	sakila
☐	sys
☐	world

Refresh 5 tables selected

Exp...	Schema Objects
☒	contacts
☒	customers
☒	mytable2
☒	mytable3
☒	new_table

Dump Structure and Dat Select Views Select Tables Unselect All

Objects to Export

☐ Dump Stored Procedures and Functions ☐ Dump Events ☐ Dump Triggers

Export Options

☐ Export to Dump Project Folder C:\Users\MUICT\Documents\dumps\Dump20241211 ...

☒ Export to Self-Contained File C:\Users\MUICT\Documents\dumps\Dump20241211 (3).sql ...

☒ Create Dump in a Single Transaction (self-contained file only) ☒ Include Create Schema

Press [Start Export] to start... Start Export

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	14:59:34	create database lab7	1 row(s) affected	0.000 sec
2	14:59:50	show databases	16 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

6.2 * .csv file_path

Syntax

```
SHOW VARIABLES LIKE "secure_file_priv";
```

Example Output

```
+-----+-----+
| Variable_name | Value |
+-----+-----+
| secure_file_priv | C:\ProgramData\MySQL\MySQL Server 8.0\Uploads\ |
+-----+-----+
1 row in set (0.00 sec)
```

For MacOS, suggest to use Lab PC (Windows) to do this job.

6.2 Import Table from *.csv

Syntax

```
LOAD DATA INFILE file_path
INTO TABLE table_name
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;
```

- **FIELDS TERMINATED BY:** This indicates that the file is a CSV (Comma-Separated Values) file.
- **ENCLOSED BY:** This is common in CSV files to handle cases where a field may contain the delimiter (,).
- **LINES TERMINATED BY:** This indicates the end of a record (row) in the CSV file.
- **IGNORE 1 ROWS:** This is useful when the first row contains headers and should not be imported as data.

6.2 Export Table to *.csv

Syntax

```
SELECT * FROM table_name  
INTO OUTFILE file_path  
FIELDS ENCLOSED BY '"'  
TERMINATED BY ','  
ESCAPED BY '"'  
LINES TERMINATED BY '\r\n';
```

- FIELDS ENCLOSED BY: Specifies that each field in the output file will be enclosed by double quotation marks (").
- TERMINATED BY: specifies that fields in the output file are separated by a comma (,).
- ESCAPED BY: Specifies the character used to escape special characters. In this case, it's a double quotation mark (").
- LINES TERMINATED BY: Specifies that each line in the output file is terminated by a carriage return (\r) followed by a newline character (\n)

Export Table to *.csv (with Column name)

Example

```
SELECT 'column1', 'column2', 'column3'  
UNION ALL  
SELECT column1, column2, column3  
INTO OUTFILE '/path/to/your/file.csv'  
FIELDS TERMINATED BY ','  
ENCLOSED BY '"'  
LINES TERMINATED BY '\n'  
FROM your_table;
```

6.2 Import/Export by Workbench of *.csv

Import

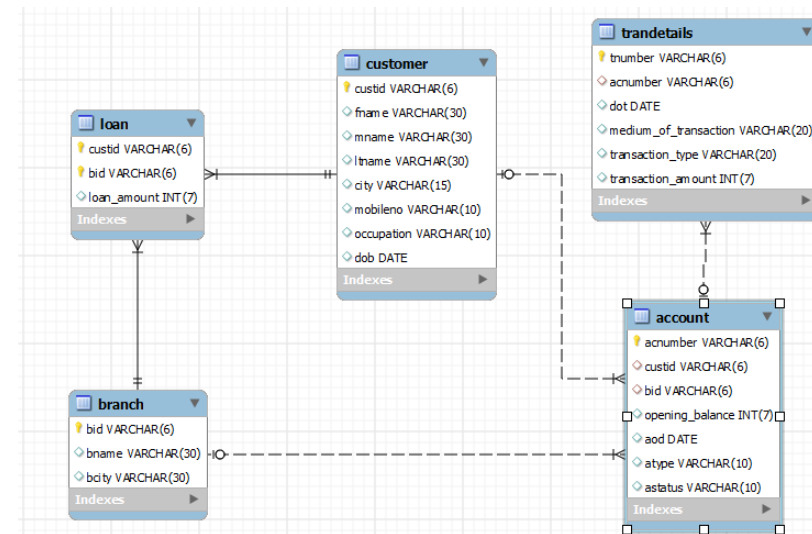
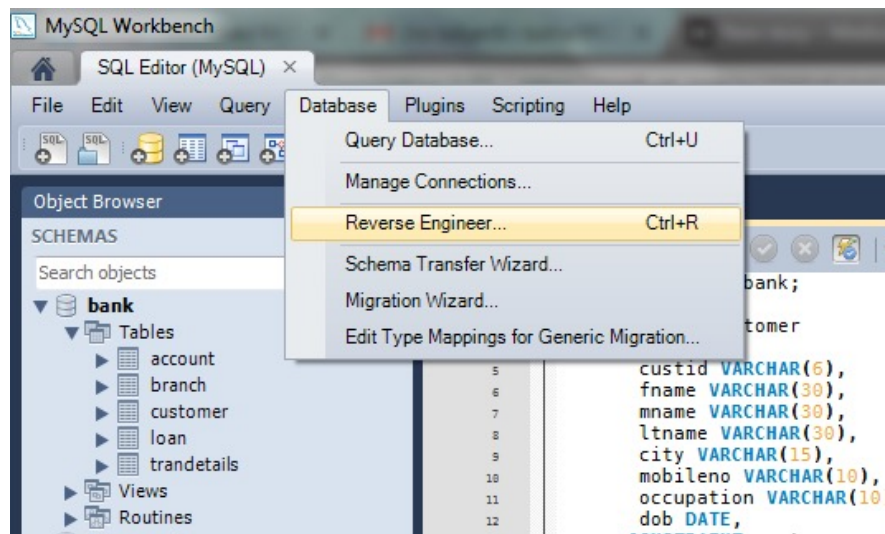
<https://www.mysqltutorial.org/mysql-basics/import-csv-file-mysql-table/>

Export

<https://www.mysqltutorial.org/mysql-basics/mysql-export-table-to-csv/>

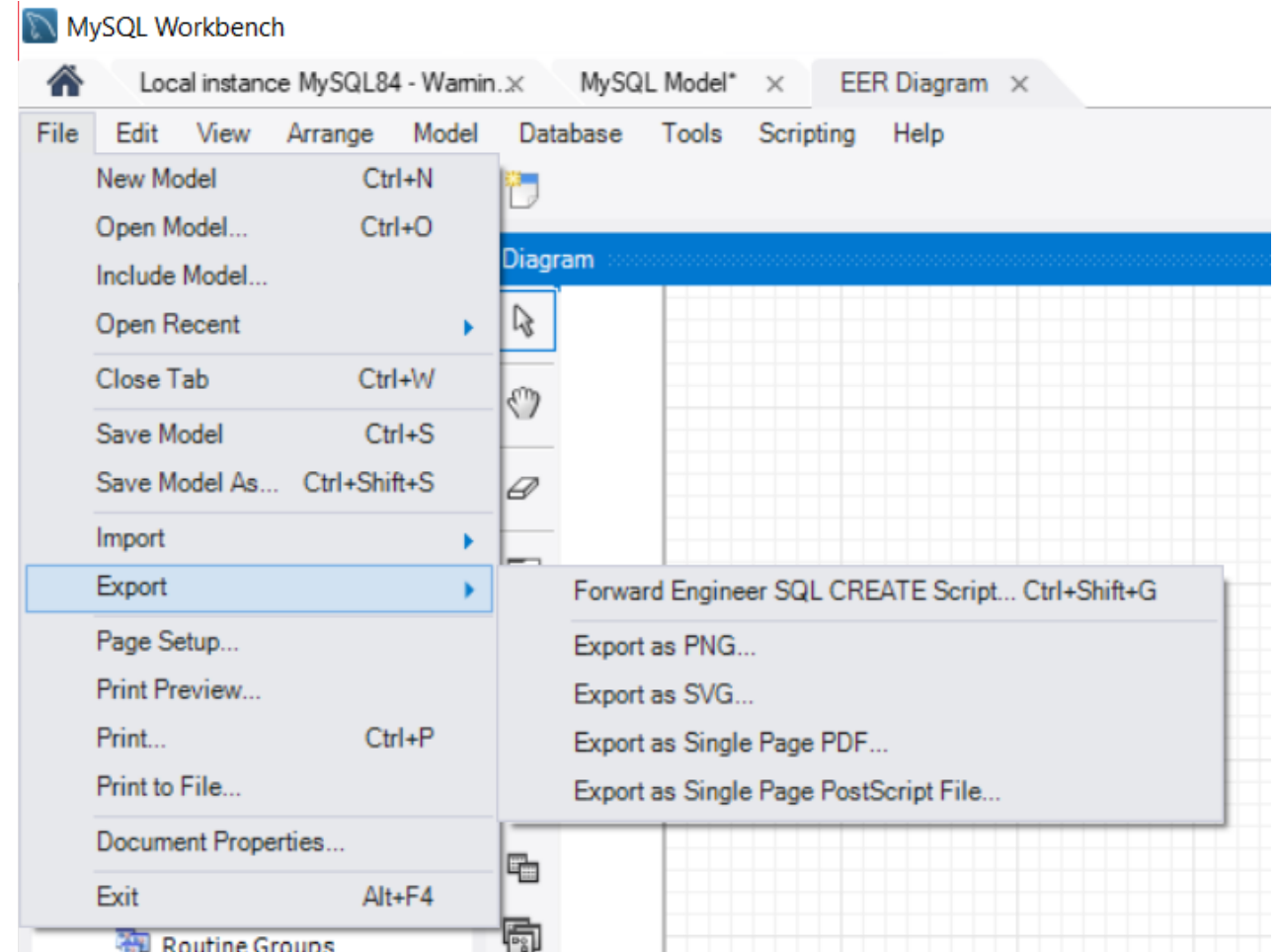
6.3 Create ER-diagram

1. First make sure you have a Database and Tables created on the MySQL server.
2. Click on **Database -> Reverse Engineer**.
3. Select your **Stored Connection** (for connecting to your MySQL Server in which database is present) from the dropdown. Then click **Next**.
4. Select your **Database** from the MySQL Server for which you want to create the ER Diagram, then click **Next**.
5. Select the **Tables** of the Database which you want to be visible on the ER Diagram, then click **Execute>**.



Save ER-diagram Image

1. Click on **File** -> **Export**.
2. Select the file type.
 - PNG for image file.
 - SVG for vector file.
 - PDF for doc file



Note. SVG is recommended for attaching to the report or poster.